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Thermal and non-thermal ablation mechanisms in crystalline silicon by femtosecond laser pulses: classical approach of the carrier density two temperature model

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Abstract

Laser micromachining has attracted considerable interest because of its wide range of applications across nearly all manufacturing sectors and mostly in semiconductors such as silicon. However, modern micro-manufacturing demands progressive product miniaturization, high accuracy, and high-precision material removal. For this purpose, a fundamental study of the interaction between ultrashort laser pulses and silicon will be valuable for studying ablation characteristics and ablation performance. The femtosecond laser pulse interaction with the silicon is divided into five parts: (a) the interaction of laser light with the carriers, (b) variation of the carrier density and carrier temperature, (c) energy exchange between carriers and lattice, (d) thermomechanical response of the material, and (e) ablation. The evolution of the carrier density and carrier-lattice energy coupling equation is solved simultaneously to determine the optimum value of the ablation width and ablation depth of femtosecond laser pulses on the silicon. The first time, 2D axial symmetry thermal and non-thermal ablation profiles were compared with the experimental result at fluence ranging from 0.75 to 9 J cm^{-2} at the wavelength of 515 nm and 180 fs laser on the silicon sample. A comparative study of damage thresholds from experiments and simulations is presented. The concordance between model calculations and experimental data demonstrates that fs laser ablation is thermal in nature in low fluence regime, whereas it is non- thermal in a high-fluence regime. Fundamental information such as the time evolution of the carrier density, carrier temperature evolution, and lattice temperature evolution can be obtained from the simulation results.

Keywords: femtosecond laser ablation, carrier density, two-temperature model, singe pulsed laser ablation of silicon

(Some figures may appear in colour only in the online journal)

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1. Introduction

In the past decades, ultrashort laser technology has attracted growing interest as a powerful manufacturing tool in various industrial fields. Ultrashort laser pulses (typically <10 ps) have high pulse intensities and enable high-resolution, high-quality micromachining of almost all solid materials with minimal thermal damage and structure sizes down to the submicron range. It is an excellent and universal tool for the very fine, direct ablative micro-structuring of solid materials. Ultrashort pulses have been demonstrated to have the potential for precise micromachining (laterally and vertically) in transparent dielectrics [1], metals [2], and semiconductors [3]. In semiconductors, silicon is the most important material, and a fundamental study between ultrashort laser pulses and silicon has been studied since the early 80s [4]. A fundamental study of the laser interaction between ultrashort laser pulses and materials is essential for controlling the topology of silicon and optimizing the process parameters. A theoretical analysis of ultrashort laser-matter interaction in a metal can be represented by the two-temperature model (TTM) which describes the temperature of the electron and lattice in non-equilibrium conditions when ultrashort laser pulses are applied [5–7]. Further, van Driel *et al* extended this model and created a carrier density n-TTM for semiconductor and dielectric materials [8]. This model was further developed based on the consideration of temperature-dependent optical properties [9]. There are many other approaches to model ultrashort laser interaction with silicon, and molecular dynamic (MD) simulation is one of the most widely used approach. MD-simulation provides a detailed morphological investigation at the atomic level, but the computation time is much longer because of the large number of atoms [10]. The hybrid model is used to determine the ultrafast thermomechanical response of silicon, where an atomic-level hybrid method couples the MDs and ultrafast two-step energy transport model [11]. It has been further extended based on the physical fundamentals of plastic deformation due to thermal stresses during silicon irradiation with ultrafast laser pulses [12]. The drift-diffusion limit of the Boltzmann-transport equation in the carrier density n-TTM leads to the thermal spike model [13] and it is further enhanced to modified potential (MOD) by using the electronic-dependent interaction potential, however, the combination of the n-TTM and MOD is not suitable for the simulation of laser ablation of bulk material as it gives the wrong ablation mechanism and different ablation depth [14]. The three temperature model (3TM) [15] was also developed to study the ultrashort laser-matter interaction because the optical phonons and acoustic phonons are not in an equilibrium state during the process; however, this effect cannot be considered in the TTM because the carrier-lattice energy relaxation time is very small and neglected as compared to the pulse duration [16].

Until now, many different studies have been conducted on the interaction of ultrashort laser pulses with silicon [8, 17]. Based on the energy transport dynamics in semiconductors, the effect of the pulse energy and pulse duration on the ablation threshold of silicon has been analysed [18, 19]. The ablation threshold also depends on the initial temperature of the material; therefore, an experimental investigation of the ablation threshold in silicon as a function of temperature at different wavelengths found that the ablation threshold reduces to 35% and 43% of the initial ablation threshold at 1030 nm and 515 nm wavelengths when the silicon temperature increased to 320 °C from room temperature, while the threshold fluence remained unchanged at 343 nm [19]. During the ultrashort laser ablation of silicon, several physical phenomena have also been observed, such as oxidation, amorphization, recrystallization, the formation of bubbles due to boiling below the surface, and finally ablation when increasing the fluence from low to threshold fluence [20]. At low fluence, Coulomb explosion [21] and high fluence plasma generation have been observed during the ultrashort laser–matter interaction of silicon [22]. When the silicon surface is exposed to ultrashort laser pulses at a fluence above the threshold fluence, ripples (pits, pillar) or laser-induced periodic surface structure have been formed [23, 24]. In the multi-pulse laser interaction of silicon, a fs-pulse train on the same location with different repetition rates is applied. Here, a quite different tendency was observed in the carrier density, lattice temperature, and electron temperature [15] because the ablation threshold is decreased by the number of pulses [20]. To reduce the effect of heat accumulation, a femtosecond burst with a THz repetition rate was used, where the temporal separation between the pulses was in the order of carrier-lattice interaction time [25, 26]. Moreover, for single pulse ablation depth determination, many pulses are applied to the material, and the crater depth was divided by the number of pulses [17] but technically involves uncertainties because of the change in the optical characteristics of the material by each pulse [27].

For short or long laser pulses in the nanosecond and microsecond range, the ablation process is dominated by heat conduction, melting, and evaporation. When an ultrashort pulse impinges on a solid, the fact that its duration is shorter than the lattice ions vibrational period means that most of the pulse energy will be transferred to the material electrons, heating them [28]. During ultrafast irradiation, it is peculiarities between the metal energy absorption in contrast to the semiconductor and dielectrics. The laser energy has been absorbed by a free electron in the conduction band in metal. When laser pulses are irradiating to the substrate in semiconductors, energetic electrons exit from valence band to conduction band due to energy transfer. In dielectric material, the bandgap between the conduction band and valence band is larger than the photon energy, nonlinear optical effects such as multiphoton transition are necessary to promote electrons from valence band to conduction band [29].

To summarize the state of the art, the carrier density n-TTM is widely used for the description of ultra-short laser–matter interaction on semiconductors. Until now, most of the simulations of ultrafast laser irradiation with semiconductors have focused on a one-dimensional approach, implemented by reasonable approximation that the laser spot size is much larger than the depth. If the laser spot size becomes smaller

than 10 μm, it is no longer valid and a multi-dimensional model should be used [29]. Mostly, a 1D model is used to determine the ablation threshold of the silicon. However, for surface functionalization applications, it is helpful to determine the ablation depth and ablation crater diameter which cannot be determined by a 1D simulation. Therefore, in this work, an axisymmetric model that combines the effect of multi-photon ionization and Auger recombination is developed to predict the spatial and temporal carrier density, carrier temperature, and lattice temperature at fluence ranging from 0.015 to 9 J cm^{-2} of 515 nm fs pulses. In this paper, we present a comparative study of simulated thermal and non-thermal ablation mechanisms with experimentally acquired structures.

2. Carrier density TTM

In semiconductors, the generation and relaxation of the electron–hole pairs play a crucial role during ultrashort pulse irradiation. If the duration of the ultrashort laser pulse is less than the carrier-lattice interaction time, the carrier and lattice systems are in non-equilibrium so that the energy transfer inside the material can be represented by the carrier density n-TTM [10, 17, 30]. Figure 1 shows a schematic diagram of ultrashort laser pulses acting on the surface of the silicon material. It is divided into five parts: absorption, heating, energy transfer, thermomechanical response of the material, and ablation. When the laser pulses irradiate the semiconductor, energy is transferred from the laser pulses to the material. This energy is absorbed by electrons and excited from the valence band to the conduction band via single-photon or multi-photon absorption depending on the laser energy and bandgap of the material, resulting in the creation of electron–hole pairs. New electron–hole pairs are generated by the energetic charge carrier via the impact ionization process, and the carrier density increases with the carrier temperature (heating). Afterward, these free electrons and holes recombine mainly via the Auger recombination process and transfer the excess energy to another free electron–hole pair. Simultaneously, the carrier system couples to the lattice system and transfers the energy from the carrier to the lattice until thermal equilibrium (energy transfer). Because of the energy transfer from the carrier to the lattice, the top surface of the material reaches the other state (thermo-mechanical response), and eventually, the temperature of the top surface reaches the vaporization temperature of the material, causing ablation (ablation).

2.1. Theoretical modelling

For the carrier energy balance equation, the spatial and temporal evolution of the carrier density can be defined as follows, where it can be affected by the laser energy absorption, impact ionization, and auger recombination [9]:

$$\frac{\partial n}{\partial t} = \left[\frac{1}{r}\frac{\partial}{\partial r}\left(D_0 r\frac{\partial N}{\partial r}\right) + \frac{\partial}{\partial z}\left(D_0\frac{\partial N}{\partial z}\right)\right] - \gamma n^3 + \eta n + \frac{\alpha I}{h\nu} + \frac{\beta I^2}{h\nu} \quad (1)$$

where n is the number of carrier density, α is the one-photon absorption coefficient, β is the two-photon absorption coefficient, I is the laser intensity, h is the Planck constant, ν is the photon frequency, γ is the auger recombination coefficient, η is the impact ionization coefficient, and D_0 is the ambipolar diffusivity which represents the electron–hole pair mobility and transport from the valence band to the conduction band.

Based on the solution of the kinetic Boltzmann equation in the relaxation time approximation, the balance equation for the carrier energy density dynamic for the electron–hole pair is written as [9, 17]:

$$C_c\frac{\partial T_c}{\partial t} = \left[\frac{1}{r}\frac{\partial}{\partial r}\left(k_c r\frac{\partial T_c}{\partial r}\right) + \frac{\partial}{\partial z}\left(k_c\frac{\partial T_c}{\partial z}\right)\right] - k(T_c - T_l) + (\alpha + \beta I + \Theta n) I - \frac{\partial n}{\partial t}\,(E_g + 3k_B T_c) - n\left(\frac{\partial E_g}{\partial n}\frac{\partial n}{\partial t} + \frac{\partial E_g}{\partial T_c}\frac{\partial T_c}{\partial t}\right). \quad (2)$$

Energy is transferred between the two subsystems of the carriers and the lattice so that the evolution of the lattice energy density can be represented as [9]:

$$[C_l + H_m\nabla(T_l - T_m)]\frac{\partial T_l}{\partial t} = \left[\frac{1}{r}\frac{\partial}{\partial r}\left(k_l r\frac{\partial T_l}{\partial r}\right) + \frac{\partial}{\partial z}\left(k_l\frac{\partial T_l}{\partial z}\right)\right] + K(T_c - T_l) \quad (3)$$

where T_c and T_l are the temperatures of carrier and lattice respectively; k_c and k_l are the thermal conductivity of the carrier and lattice respectively, C_c and C_l are the heat capacity of the carrier and lattice respectively, K_B is the Boltzmann constant, t is the time, Θ is the carrier absorption coefficient, E_g is the bandgap energy of silicon, H_m is the latent heat of melting, and K is a coupling constant. The phase transition of the solid to the gaseous phase occurs when the lattice temperature for the evaporation reaches sufficient energy to overcome the latent heat of fusion so that the lattice equation has to be modified properly to include the latent heat of melting. A suitable representation of the ∇ function should be applied to smooth transition [12]

$$\nabla(T_l - T_m) = \frac{1}{\sqrt{2\pi}\Delta}\mathrm{e}^{-\left[\frac{(T_l - T_m)^2}{2\Delta^2}\right]} \quad (4)$$

where Δ is the range of 10 K–200 K, depending on the temperature gradient.

The attenuation of the laser beam along the propagation direction z can be defined by a 1D ordinary differential equation [9]:

$$\frac{\partial I(r,t,z)}{\partial z} = -(\alpha + \beta I(r,t,z) + \Theta n) I(r,t,z). \quad (5)$$

After the integration, the laser intensity in the surface can be modelled as [31]:

$$I = \frac{(\alpha + \Theta n) I_0\, e^{-(\alpha+\Theta n)z}}{(\alpha + \Theta n) + \beta I_0\left[1 - e^{-(\alpha+\Theta n)z}\right]} \quad (6)$$

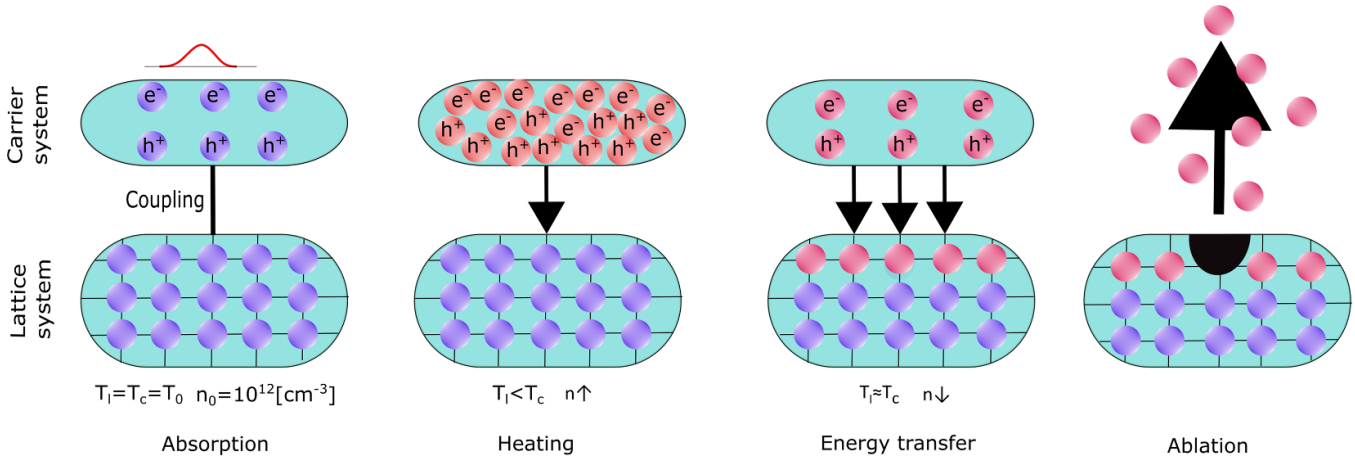

Figure 1. Schematic diagram of ultrashort pulse laser interaction with semiconductor. Where T_l and T_c is the temperature of lattice and carrier and n_0 is the initial carrier density of silicon.

Table 1. Thermophysical properties of silicon used in this study.

Parameter (reference)	Nomenclature	Value (unit)
Liner absorption coefficient [19]	A	$5.02 \times 10^5 \times \exp(T_i/430)$ (1 m^{-1})
Auger recombination coefficient [33]	Γ	3.8×10^{-43} (m^6 s^{-1})
Impact ionisation coefficient [35]	η	$3.6 \times 10^{10} \times e^{-\frac{1.5E_g}{K_b T_i}}$
Bandgap of material [9]	E_g	$1.16–7.2 \times 10^{-4}[T_i^2/(T_i + 1080\ (K))] - 1.5 \times 10^8 n^{1/3}$ (eV)
Boltzmann constant	K_b	1.38×10^{-23} (J K^{-1})
Heat capacity of carrier [11]	C_c	$3nK_b$ (J m^{-3} K^{-1})
Thermal conductivity of carrier [36]	k_c	$-3.47 \times 10^{-8} + 4.45 \times 10^6$ Te (eV/sÅK)
Carrier lattice coupling constant [9]	K	$3nK_b/\tau_{c\text{-}l}$ (W m^{-3} K^{-1})
Carrier-lattice relaxation time [34]	$\tau_{c\text{-}l}$	$240\{1 + (n/6 \times 10^{26})^2\}$ (fs)
Heat capacity of lattice [36]	C_l	$1.987 \times 10^6 + 354T_i - 3.68 \times 10^6 T_i^{-2}$ (J m^{-3} K^{-1})
Thermal conductivity of lattice [37]	k_l	$1.585 \times 10^5 T_i^{-1.23}$ (W mK^{-1})
Reflectivity [38]	R	$0.37 + 10^{-5} \times (T_i - T_{room})$
Ambipolar diffusivity [33]	D_0	$1.8 \times 10^{-3}(300/T_i)$ (m^2 s^{-1})
Vaporization temperature [39]	T_{vap}	3514 (K)
Latent heat of fusion [39]	H_m	1.797×10^6 (J kg^{-1})
Critical density [19]	N_{cr}	2.7×10^{27} (1 m^{-3}) at 515 nm
Melting temperature [39]	T_m	1689 (K)

where I_0 is the laser intensity at the top surface ($z = 0$). The laser beam is Gaussian, and the temporal and spatial evolution of the laser intensity can be expressed as [9]:

$$I_0(r,t,z=0) = \sqrt{\frac{4ln2}{\Pi}}\left(\frac{F}{\tau_p}\right)\exp\left(-2\frac{r^2}{{w_0}^2}\right) \times (1-R)\exp\left(-4ln2\frac{(t-t_0)^2}{{\tau_p}^2}\right) \quad (7)$$

where R represents the surface reflectivity, F is the laser fluence, τ_p is the pulse width, w_0 is the beam radius at the laser-focal plane, and the pulse peak arrives at t_0.

To estimate the ablation crater depth and radius, the evolution of the carrier density and carrier-lattice energy coupling equation was solved simultaneously. The temperature-dependent thermophysical properties used in the simulations are listed in table 1. Several groups used different values for the parameter, such as the carrier-lattice relaxation time ($\tau_{c\text{-}l}$) used between 1 and 10 ps [23, 32, 33], but based on the experiment on silicon, it is $240 \times [1 + (N_c/6 \times 10^{26})^2]$fs [34]. Most of the parameters used in the simulation closely resembled our experimental conditions. Moreover, for all simulations, the two-photon absorption coefficient is neglected because the photon energy is greater than the bandgap of silicon [11].

2.2. Experimental work and methodology

The laser ablation experiments were performed on polished crystalline silicon (orientation <100>) with a thickness of 500 μm. A femtosecond Yb:KGW laser system (Pharos from Light Conversion Ltd) was used for the experiment, which provided pulses of 180 fs duration at a maximum repetition rate of 600 kHz and maximum average power of 6 W at 1030 nm. For the ablation experiments, we used a second harmonic at 515 nm. The sample was placed on a motorized linear *XYZ* stage, and the laser was focused on the sample

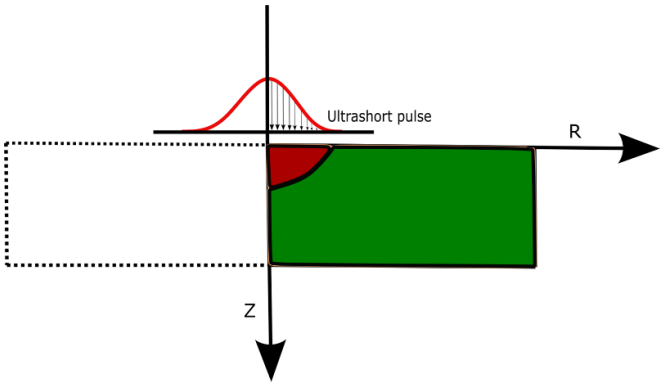

Figure 2. 2D-axisymmetric model used in the simulation.

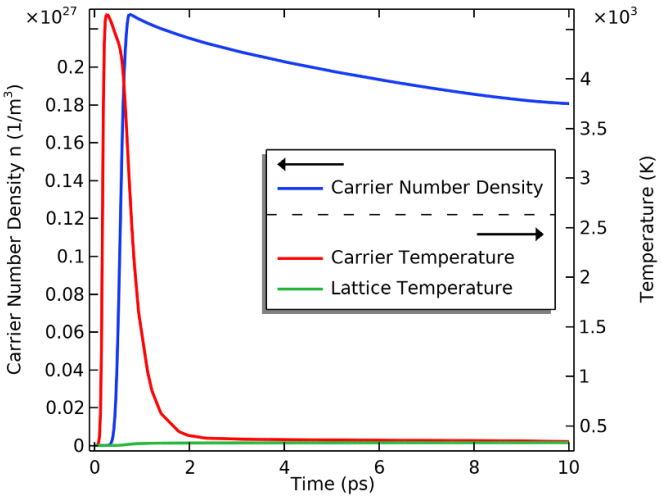

Figure 3. Time evolution of carrier density (n), carrier temperature (T_c) and lattice temperature (T_i) at a centre ($r = 0, z = 0$) of silicon sample irradiated by the 180 fs laser with the wavelength of 515 nm at a fluence $F = 0.015$ J cm^{-2}.

using an F-theta lense with a 100 mm working distance. The laser setup is optimized to produce a spot size (df) of approximately 20 μm.

Due to the rotational symmetry of the laser ablation process, a 2D-axisymmetric model has been developed which has 30 μm in a radial direction and 5 μm in an axial direction, representing half of the target material as shown in figure 2. The ultrashort laser pulse with gaussian distribution was applied to the top surface and propagated in the z-direction. The carrier density TTM was solved with an initial ambient temperature of 300 K and ambient pressure of 1 bar. An initial carrier density of 10^{12} cm^{-3} corresponding to silicon is used in this model [8, 9, 38].

2.3. Ablation mechanisms

There are two possible ablation mechanisms used in ultrashort laser ablation in silicon: thermal and non-thermal. Thermal ablation occurs if the lattice temperature reaches the vaporization temperature [17]. Non-thermal ablation occurs when the carrier density reaches a critical density [19]. The ablated crater profile is obtained by either removing the elements that reach the vaporization temperature or reach the critical density of the material. The other phenomenon is termed as phase explosion mechanism for the description; at a higher fluence (>1 J cm^{-2}) the material goes to the superheating regime at a critical temperature and it is assumed that heterogeneous nucleation occurs due to phase explosion when the lattice temperature reaches the $0.9T_c$ (critical temperature) [40–42].

3. Result and discussion

3.1. Carrier number density, carrier and lattice temperature distribution

The main objective of this work is to predict ablation threshold and topographic morphology of single-shot laser ablation and compare it with the experimental results. Figure 3 depicts the evolution of the carrier number density, the carrier and lattice temperature of the crystalline silicon at the centre surface point ($r = 0, z = 0$) heated by a single pulse with a laser fluence of 0.015 J cm^{-2}. It is important to note that peak carrier temperature occurs at $t = 0.4$ ps and 4500 K, however, the peak carrier density reaches 2.25 $\times$ 10^{26} m^{-3} around 1.2 ps at a much later time as compared to the carrier temperature. The reason behind the earlier peaking for carrier temperature is as follows. As can be seen in equation (2), five energy sources are responsible for elevated carrier temperature. In the beginning, absorbed laser energy heat source and heat source related to change in carried density outweigh the other three [9]. The resulting net energy change rate is positive, albeit small. As a result, only a small number of electrons–hole pairs are created during this early stage that leads to a small carrier heat capacity ($3nK_b$). As a consequence, a rapid and noticeable rise in carrier temperature. As more laser energy has been absorbed over time, the carrier density increases dramatically.

This is due to the fact that the heat source of the carrier temperature and the energy change needs that time to uplift carrier density. Similarly, numerical results show that at an earlier stage, a very small number of electron–hole pairs are created and as time increases, the carrier number density increases drastically due to laser light excitation of electron from valance band to conduction band by single photon absorption. Afterward, it declines because of the Auger recombination process. Simultaneously, after reaching a peak, the carrier temperature continues falling as time increase because of the energy exchange from the carrier to the lattice. As time passes, the carrier and lattice temperatures are in equilibrium at 8 ps through the process of electron-lattice energy relaxation.

Figure 4 illustrates the evolution of the carrier density, carrier temperature, and lattice temperature at a fluence 0.15 J cm^{-2}. One can notice that high-density carriers are excited from the valence band to the conduction band, and reach a peak value of 2.1 $\times$ 10^{27} m^{-3} in 0.61 ps due to one-photon absorption and the 2nd peak is the carrier temperature at 0.65 ps which reach a maximum value of 9600 K. The rapid decrease in the carrier density is mainly owing to the auger

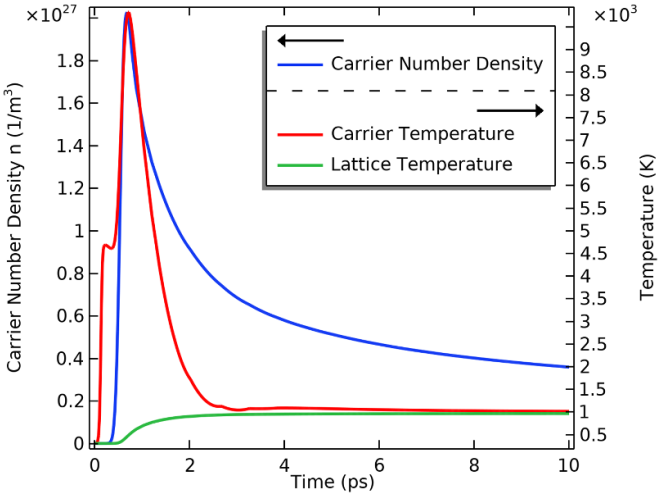

Figure 4. Time evolution of carrier density (n), carrier temperature (T_c) and lattice temperature (T_l) at a centre ($r = 0$, $z = 0$) of silicon sample irradiated by the 180 fs laser with the wavelength of 515 nm at a fluence $F = 0.15$ J cm^{-2}.

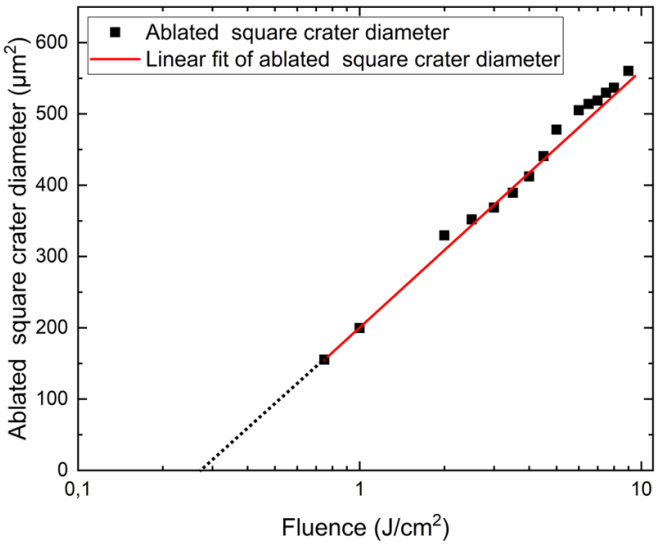

Figure 5. Squared diameter of the region ablated by 180 fs ultrashort pulses on the surface of silicon as a function of the pulse fluence.

recombination process where excess energy from the electron–hole recombination is transferred to electrons or holes that are subsequently excited to higher energy states within the same band so that afterward net charge carrier density remains constant. The carrier density should decrease the n^3 decaying rate because of the Auger recombination effect. The carrier density reached its maximum slightly earlier than the carrier temperature after the laser pulse. Unlike in the previous case at very low fluence where the carrier temperature peaks in the very beginning. As can be seen in figure 4, the carrier temperature rises sharply but declines slightly at $t = 0.2$ ps and again upsurges to reach the maximum value. The impact ionization effect causes a minor drop in carrier temperature because the energetic charge carrier can lose energy by producing other charge carriers; subsequently, the carrier density rises dramatically.

3.2. Determination of the threshold fluence

The threshold fluence indicates that a minimal laser fluence or laser energy is irradiated to remove material from the substrate. The ablation threshold fluence values depend on several laser parameters (pulse duration, wavelength, and spot size) and material properties. In 1982, Liu *et al* developed a method for measuring threshold fluence [43]. With a laser beam having a Gaussian spatial distribution, a square of ablated crater diameters is plotted as a function of the logarithmic pulse fluence for the determination of the threshold fluence [43], following the classical expression:

$$D^2 = 2 \cdot {\omega_0}^2 \cdot \ln\left(\frac{F_0}{F_{\text{th}}}\right) \tag{8}$$

where ω_0 is the beam spot size at the sample surface, F_{th} is the ablation threshold fluence, and F_0 is the fluence.

The linear fitting of the squared diameter (D^2) versus the logarithmic pulse fluence (F_0) range from 0.75 to 9 J cm^{-2} as shown in figure 5. The ablation crater diameter was measured using an optical microscope (Zeiss 'Axio Scope.A2' Vario). For the ablation process, the threshold fluence of 0.27 ± 0.02 J cm^{-2} is determined by interception of the regression curve. Figure 6 shows the carrier density, carrier temperature, and lattice temperature at the Si surface irradiated by a 180 fs laser pulse over a range slightly below the threshold fluence. The carrier density rapidly increases when laser fluence is changed from 0.19 to 0.25 J cm^{-2}, and it approached to saturation by reaching the critical density of 2.7×10^{27} m^{-3} at a fluence of 0.22 J cm^{-2}. In parallel, the carrier temperature and lattice temperature are also rising with increasing fluence, and after few a picosecond, both are in equilibrium. One can notice that at fluence of 0.25 J cm^{-2}, the lattice temperature exceeds the melting temperature of silicon.

A comparison between the damage threshold based on the ablation process and the simulation process is shown in table 2. It shows good agreement between simulation and experimental results; however, the condition defining melting fluence threshold was indeed closer to the experimental data.

3.3. Ablated crater profile on silicon

In order to explore the dependence of fluence on the ablation morphology, a theoretical framework relying on the carrier density two-temperature approach is utilized. During ultrafast laser heating in semiconductors, the carrier number density is not constant spatially or temporally because of the Auger recombination process. Thus, once the carrier and lattice are in equilibrium, the carrier number density field is extracted.

Figure 7 illustrates the comparison between the simulated thermal ablation and non-thermal ablation profiles. Due to potential emergence in both energy density and heat absorption, the ablation depth, and the ablation crater radius climbed up as fluence continued to increase from 2 to 9 J cm^{-2} in both

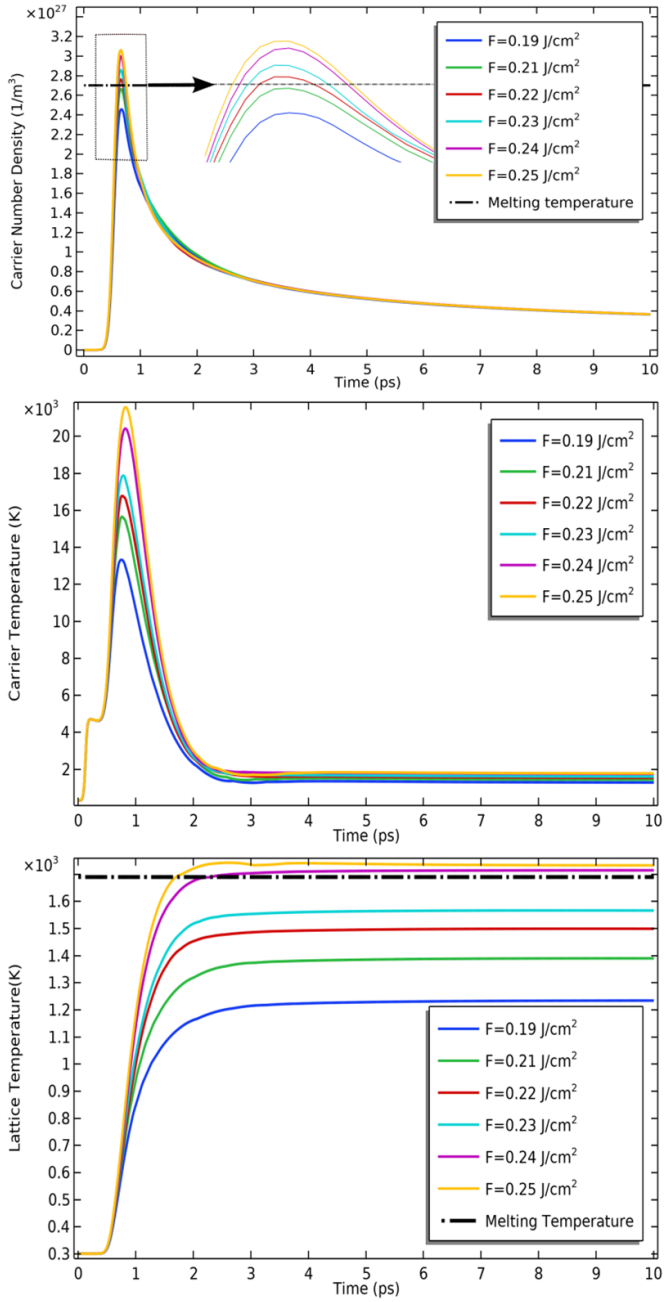

Figure 6. The evolution of carrier number density, carrier temperature and lattice temperature at silicon surface irradiated with a laser pulse of 180 fs from 0.19 to 0.25 J cm^{-2} fluence.

Table 2. Single pulse laser threshold of silicon at 515 nm wavelength and 180 fs pulse duration.

Process	Threshold fluence
Ablation (experiment)	0.27 ± 0.02 J cm^{-2}
Critical carrier density (simulation)	0.22 J cm^{-2}
Melting temperature (simulation)	0.25 J cm^{-2}

thermal and non-thermal simulated ablation profiles. Compared to longer laser pulses, the femtosecond laser ablation mechanism is different because of the extremely short pulse width and high energy density per unit pulse. Generally, the carrier-lattice relaxation time ($\tau_{c\text{-}l}$) in silicon is 1–10 ps [23, 32, 33]. Therefore, the mechanism of femtosecond laser ablation differs from that of longer pulsed laser ablation. As the 180 fs pulse duration is shorter than the electron-lattice relaxation time, the energy deposited on the material is faster than the electron-lattice relaxation time resulting in a low heat affected zone (<1 μm) after femtosecond laser ablation. Figure 8 illustrates the comparison of the measured ablation profile, the simulated lattice temperature fields (light yellow-black curve), and the simulated carrier number density field (red-blue curve) after the electron and lattice temperatures reached equilibrium at a fluence of 9 J cm^{-2}. As can be seen, a narrow heat-affected zone between the outer and inner surfaces was observed in the simulated profile. As shown in figure 8, the calculated temperature range between vaporization, and melting temperature is relatively small, and the strong decreasing gradient to room temperature displays a relatively small heat-affected zone and minimal heat diffusion. Similarly, the surrounding carrier density (left top of figure 8) does not reach the critical density at the same relative location from the pulse centre. Again, no further decomposition or re-solidification of the material occurred.

The ablation crater profiles were obtained from the axisymmetric numerical simulation with laser fluence ranging from 0.75 to 9 J cm^{-2} with thermal and non-thermal ablation mechanisms after the carrier and lattice were in equilibrium. A comparison of the ablated crater diameter and depth for different laser fluences with the simulated profile is shown in figures 9 and 10. The black square represents the experimental result, red circles represent the thermal simulation ablation result, and the blue triangle shows the non-thermal simulation results. The ablation depth was measured using profilometry and FIB (focused ion beam) SEM cross sections. Due to uncertainties of profilometer stylus tip radius and observed deviations of the depth value between both methods, we reference the data to the FIB measurements.

The simulated thermal ablation depth and diameter were slightly larger than the simulated non-thermal ablation depths and diameters. The simulated (thermal and non-thermal) ablation depths are overestimated compared to the experimental results. However, the only substantial difference was observed compared with the non-thermal simulation and experimental results. Overall, the three data lines (thermal ablation, non-thermal ablation, and experiential results) follow the similar trend. When a femtosecond laser pulse is applied to the silicon for the 1st few picoseconds a non-equilibrium excitation state inside the crystal is created. Subsequently, the carrier and lattice energy exchange leads to an equilibrium state. In the low fluence regime ($\leqslant$1 J cm^{-2}), the simulated thermal and non-thermal profiles exhibit very high similarity in shape and size; however, if the fluence is higher than the 1 J cm^{-2}, non-thermal ablation gives closer results to the experiment because the laser–matter interaction time is less than the characteristic carrier-lattice relaxation times. In the low fluence regime, ablation occurred mainly due to the optical penetration, which

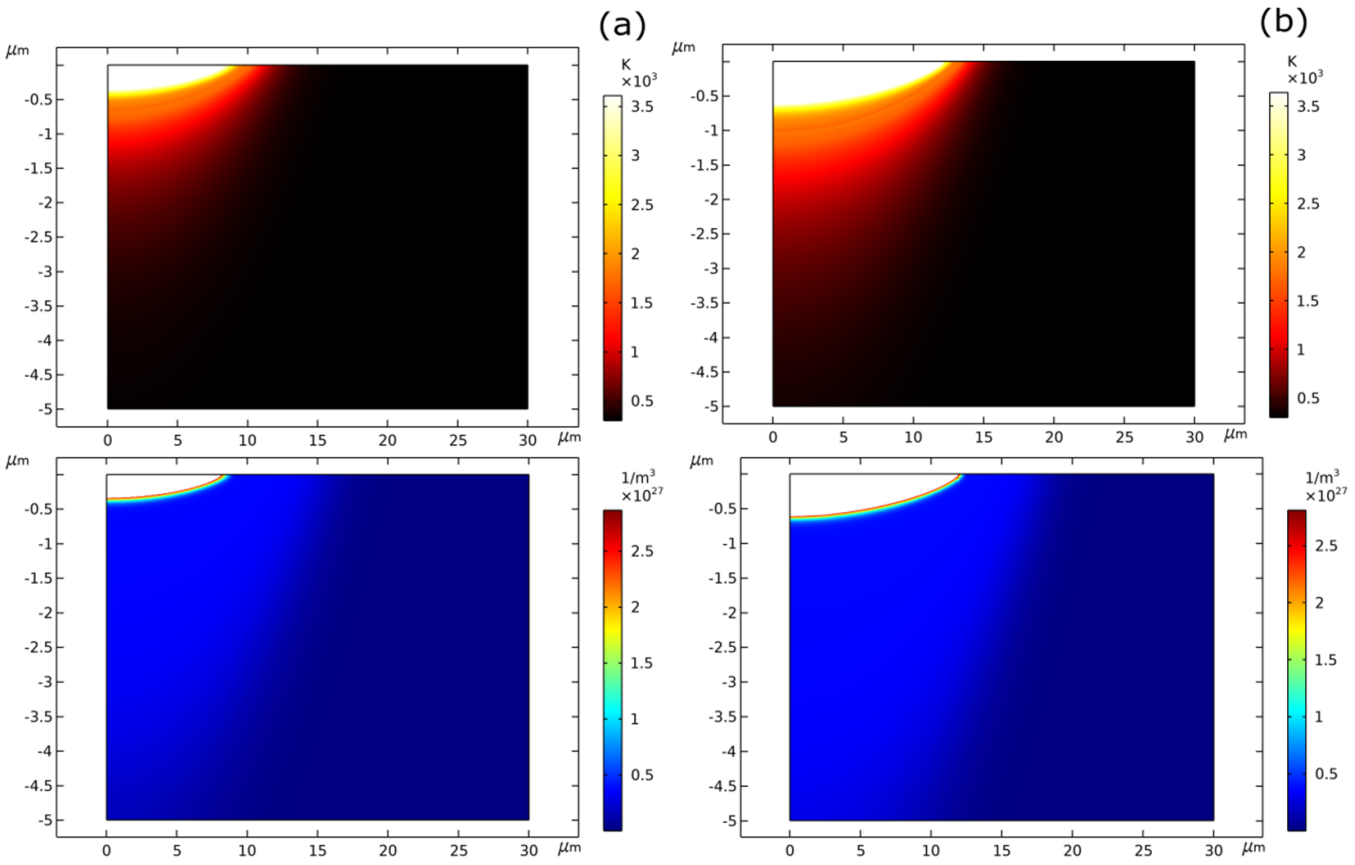

Figure 7. Axi-symmetric simulation results of ablated crater profiles in dependence of carrier density (red-blue curve) and lattice temperature (light yellow black curve) by a pulse duration of 180 fs at fluence of (a) 2 J cm^{-2} and (b) 9 J cm^{-2}.

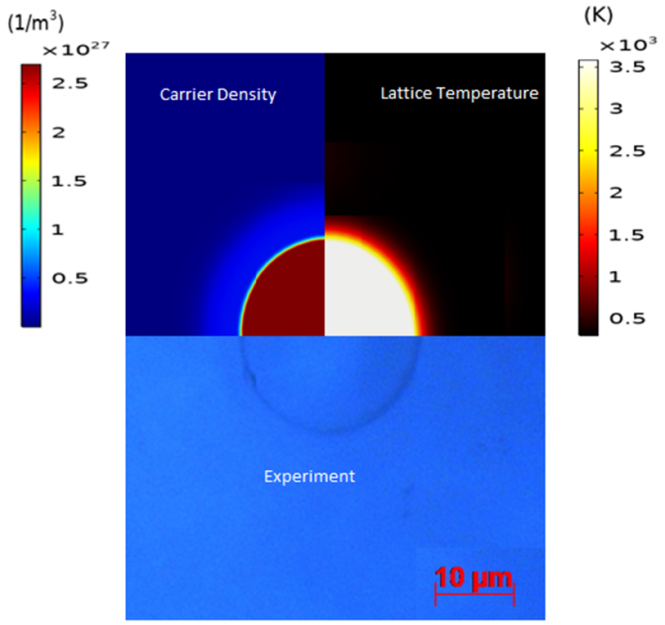

Figure 8. Experimental and numerical simulation results.

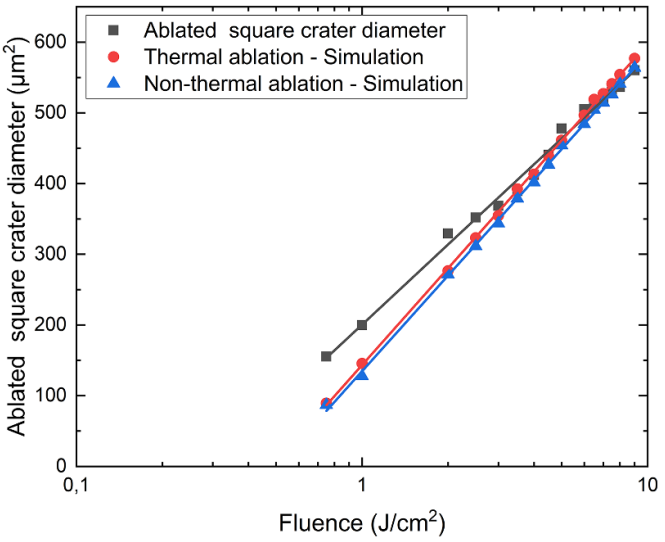

Figure 9. Comparison between thermal, non-thermal and experimental square crater diameter of silicon at wavelength of 515 nm and 180 fs pulse duration.

is equal to ablation depth. As the fluence continues to increase, energy transfer extends beyond the optical penetration depth, necessitating the consideration of the carrier's thermal diffusivity [32]. Moser *et al* perceives that the overestimation of ablation depth with thermal ablation (3514 K) in middle fluence regime (1–3.8 J cm^{-2}) similitude to our study and predict that transgression of temperatures over 3800 K seems to be a good simplified ablation condition to estimate ablation crater [17]. In femtosecond laser interaction with silicon, sufficient

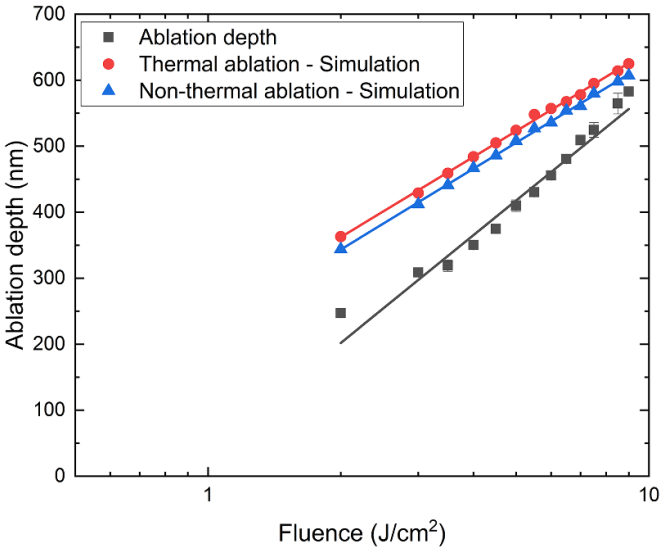

Figure 10. Comparison between thermal, non-thermal and experimental ablation depth of silicon at wavelength of 515 nm and 180 fs pulse duration.

energy has been provided so that material remains in local critical phases after the recombination process and non-thermal ablation occurs. Thus, the deviation from experimental and simulation results gives rise to the conclusion that low fluence regime thermal ablation is dominant, and in the high fluence regime non-thermal ablation is prevalent to understand the ablation characteristics of silicon with fs pulses.

4. Conclusion

This work shown for the first time temporal and spatial evolution of the carrier density, carrier temperature, and lattice temperature in 2D provided based on the Boltzmann transport equation after fs-irradiation. Moreover, the influence of the pulse energy on the carrier number density, carrier temperature, and lattice temperature is discussed. Because carriers are not stimulated from the valence band to the conduction band at very low fluence, the peak of the carrier temperature occurs early. However, at moderate or high fluence, the carrier density reaches its peak as soon as the laser pulse is applied. The damage thresholds are calculated based on the critical carrier density and crystal melting temperature which is in good agreement with the experimental data. However, the melting threshold gives an identical result with experimental because for low fluence melting is a purely thermal process, whereas, for high fluence, it is a non-thermal process. Additionally, the effect of laser fluence on the morphology of the micro-crater and ablation yield was investigated. For the first time, the simulated thermal and non-thermal squared crater diameter and depth for the axis-symmetric model compared with experimental values with different fluences, which demonstrates that the non-thermal ablation mechanism gives more relevant results because the pulse duration is much shorter than the relaxation time. The femtosecond laser ablation process on silicon can be described in terms of a very simplified thermal ablation mechanism in a low fluence regime and a non-thermal ablation mechanism in a high fluence regime. As the laser fluence rises, the ablation crater formation and ablation depth increase because more laser energy is absorbed. It has been still an open question to determine the critical density in semiconductors, although it has a range of 10^{27} to 10^{28} m^{-3} depending on the material and wavelength. Finally, it concludes that ablation morphology and ablation threshold can be predicted by using the carrier density TTM with relevant laser parameters.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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